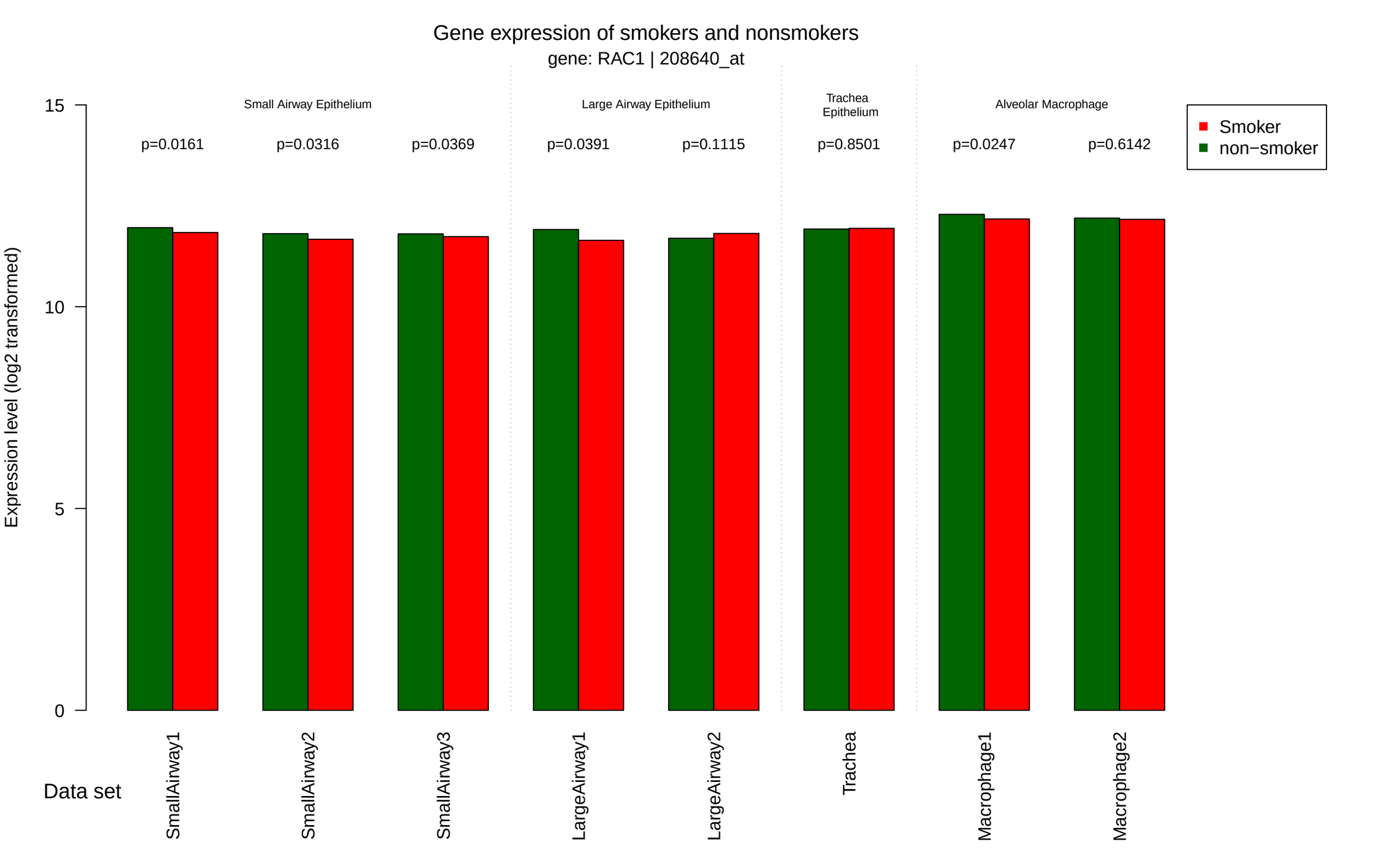
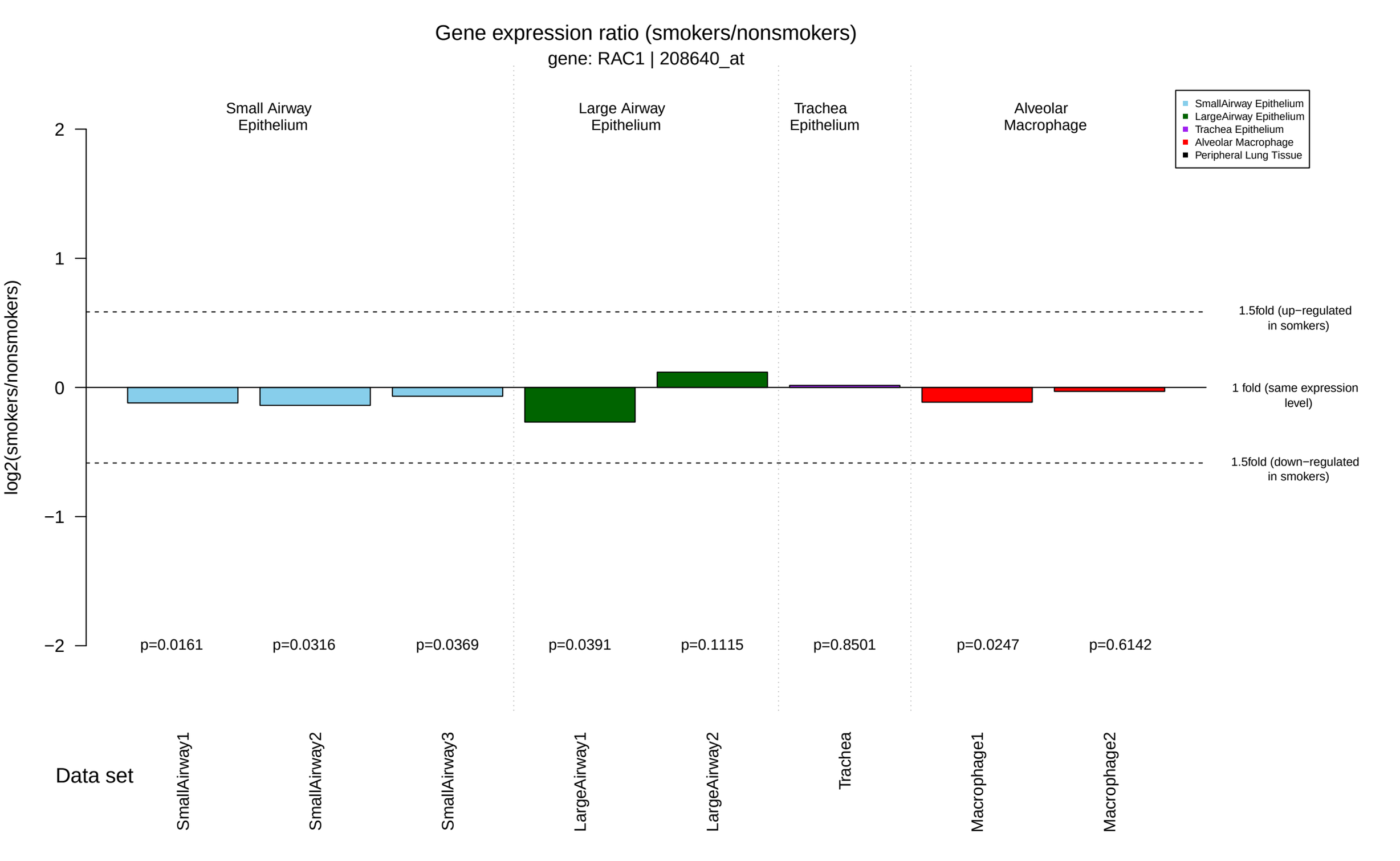
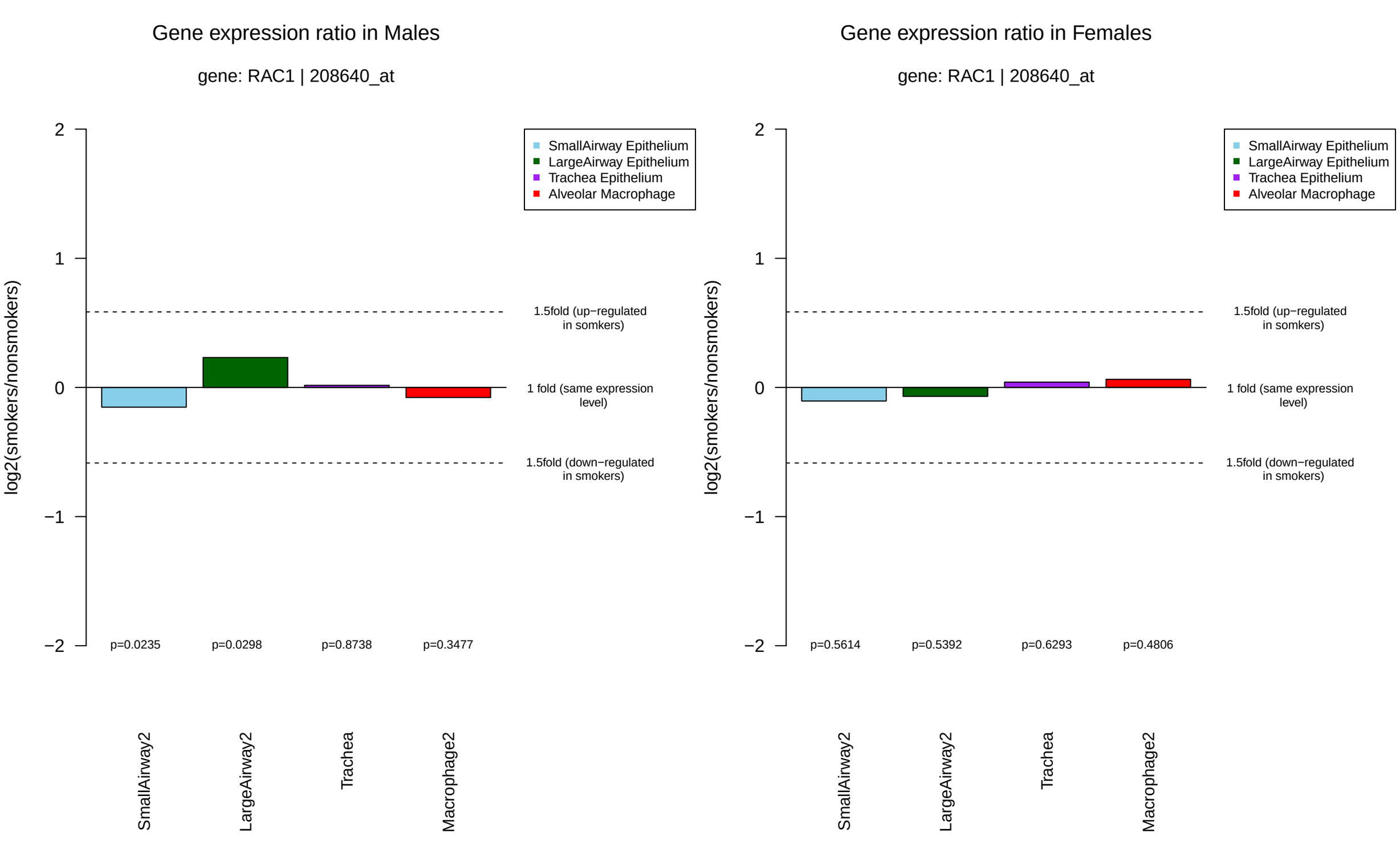
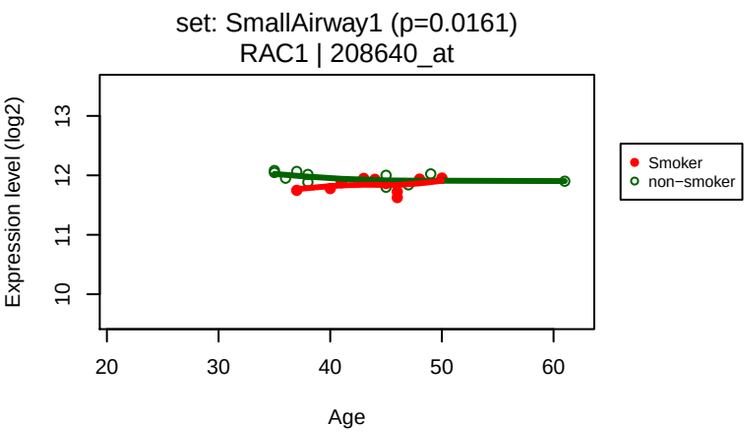
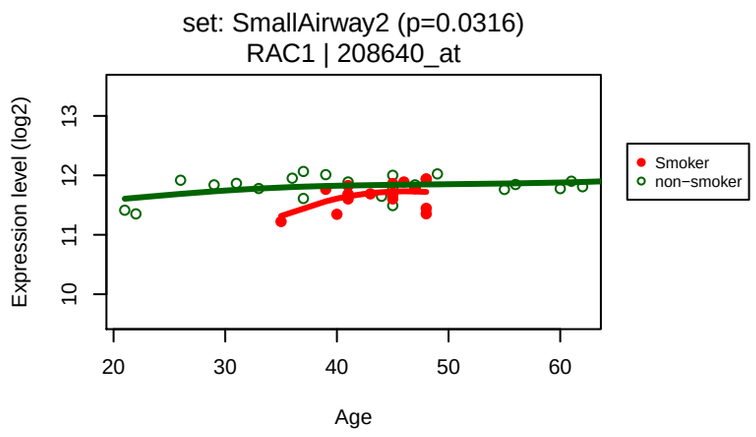
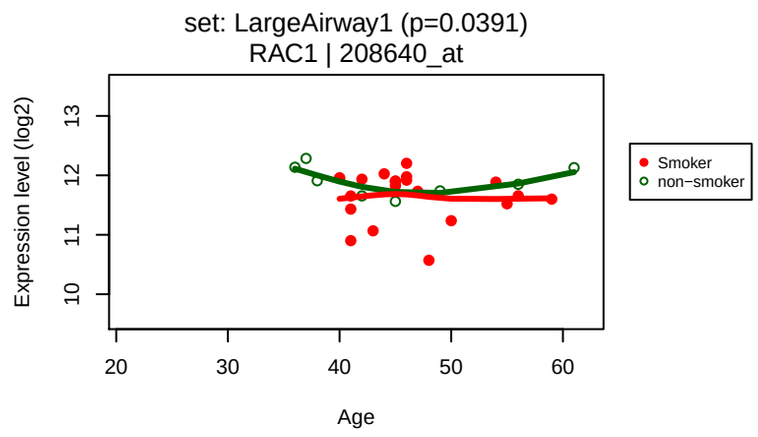

Gene expression of smokers and nonsmokers
gene: RAC1 | 208640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0161
p=0.0316
p=0.0369
p=0.0391
p=0.1115
p=0.8501
p=0.0247
p=0.6142
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

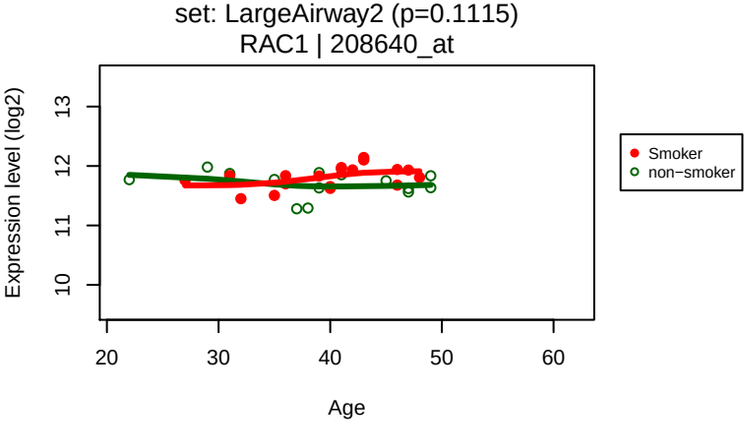

Gene expression ratio (smokers/nonsmokers)
gene: RAC1 | 208640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0161
p=0.0316
p=0.0369
p=0.0391
p=0.1115
p=0.8501
p=0.0247
p=0.6142
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

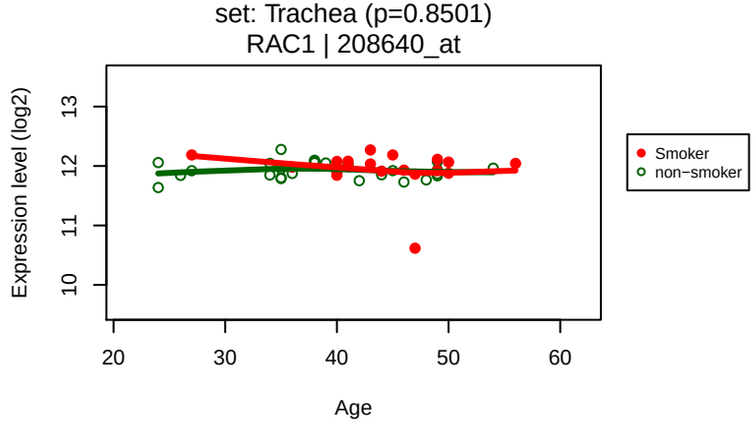

Gene expression ratio in Males
gene: RAC1 | 208640_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0235
p=0.0298
p=0.8738
p=0.3477
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAC1 | 208640_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5614
p=0.5392
p=0.6293
p=0.4806
SmallAirway2
LargeAirway2
Trachea
Macrophage2

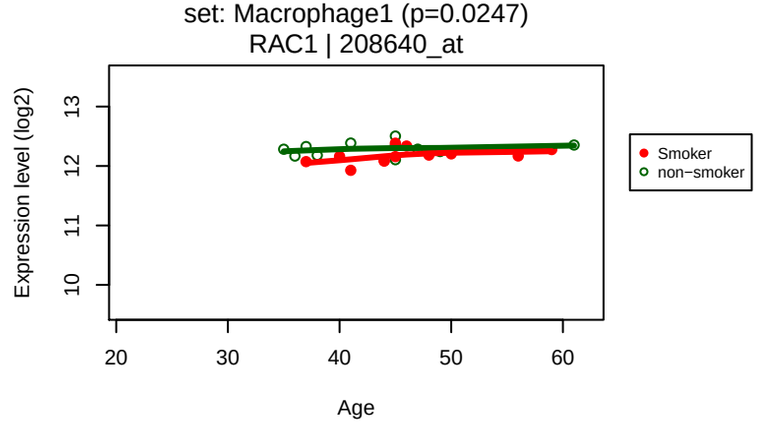

set: SmallAirway1 (p=0.0161)
RAC1 | 208640_at
Expression level (log2)
Age
Smoker
non−smoker

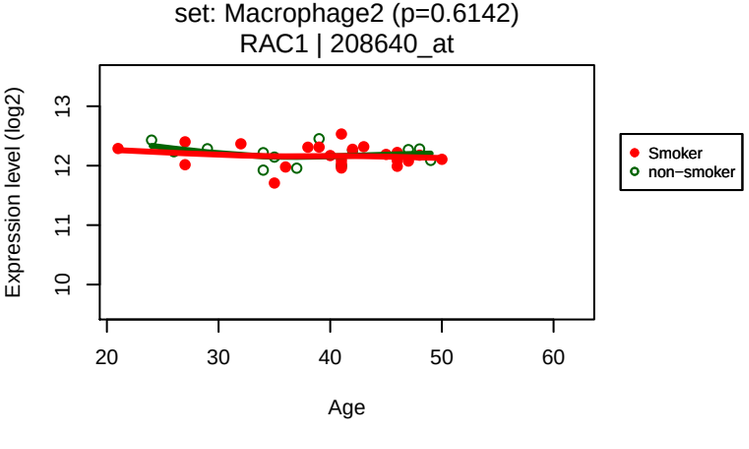

set: SmallAirway2 (p=0.0316)
RAC1 | 208640_at
Expression level (log2)
Age
Smoker
non−smoker

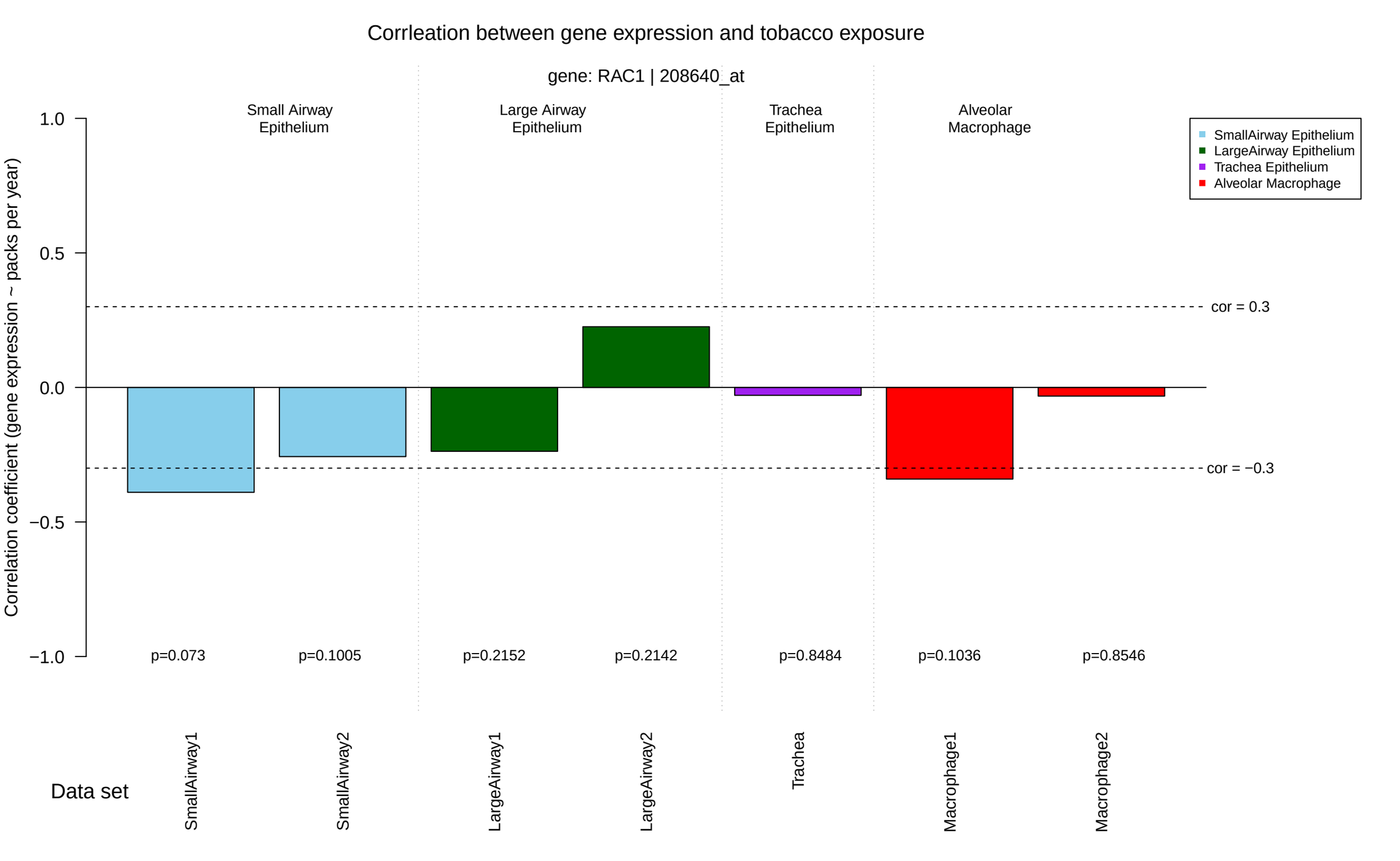

set: LargeAirway1 (p=0.0391)
RAC1 | 208640_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1115)
RAC1 | 208640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8501)
RAC1 | 208640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0247)
RAC1 | 208640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6142)
RAC1 | 208640_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAC1 | 208640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.073
p=0.1005
p=0.2152
p=0.2142
p=0.8484
p=0.1036
p=0.8546
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2